СОВКОМБАНК

Gen AI implementation for increase of operational efficiency

About us



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Examples of successful Gen AI implementation are related to document processing, AI-consultants in specific areas

1. Registration and routing of incoming documents

2. Preparation of responses to various types of external requests

3. AI-consultants in specific areas

4. SQL-generator (Text to SQL)

Document processing: reconstruction of the process requires both AI and algorithmic robotization



Document

Document processing: reconstruction of the process requires both AI and algorithmic robotization



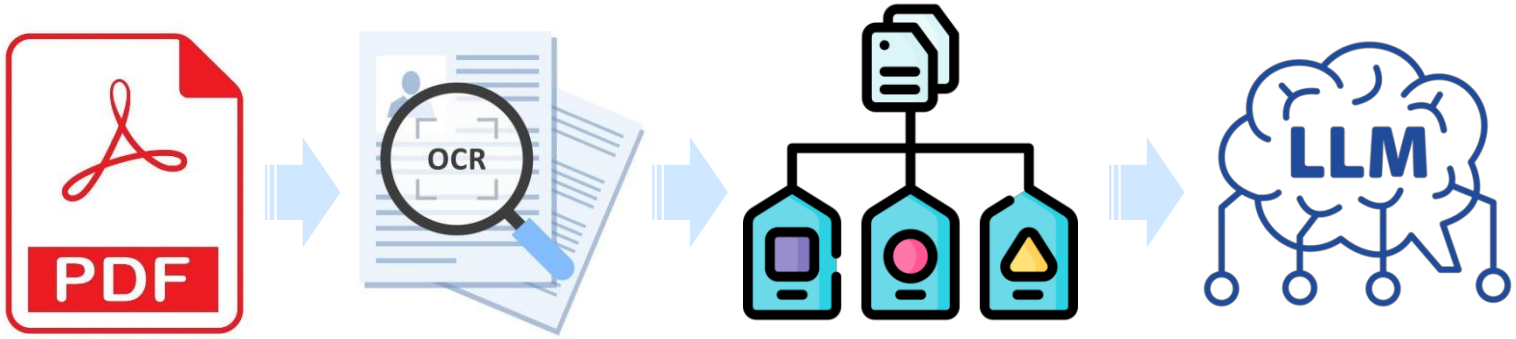
Document OCR text layer

Document processing: reconstruction of the process requires both AI and algorithmic robotization



Document OCR text layer Classification

Document processing: reconstruction of the process requires both AI and algorithmic robotization



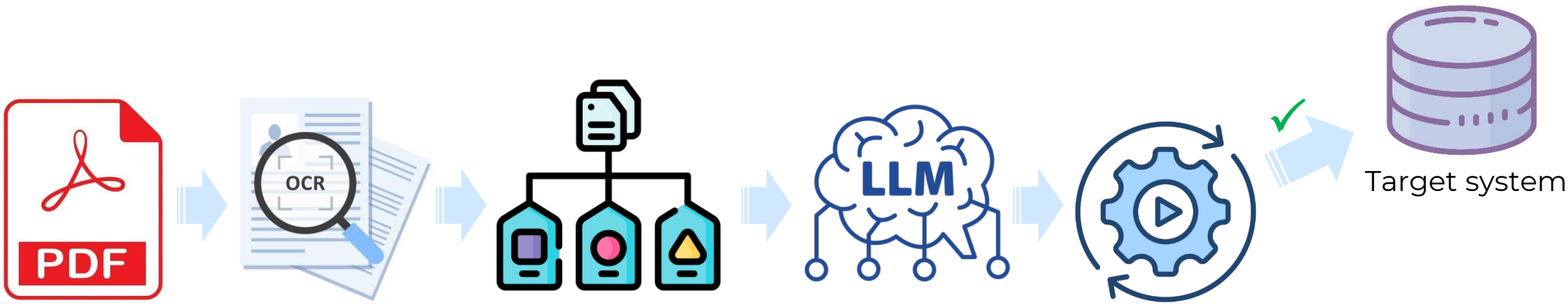
Document | OCR text layer | Classification | Extraction of attributes

Document processing: reconstruction of the process requires both AI and algorithmic robotization



Document | OCR text layer | Classification | Extraction of attributes | Automatic validation

Document processing: reconstruction of the process requires both AI and algorithmic robotization



Document

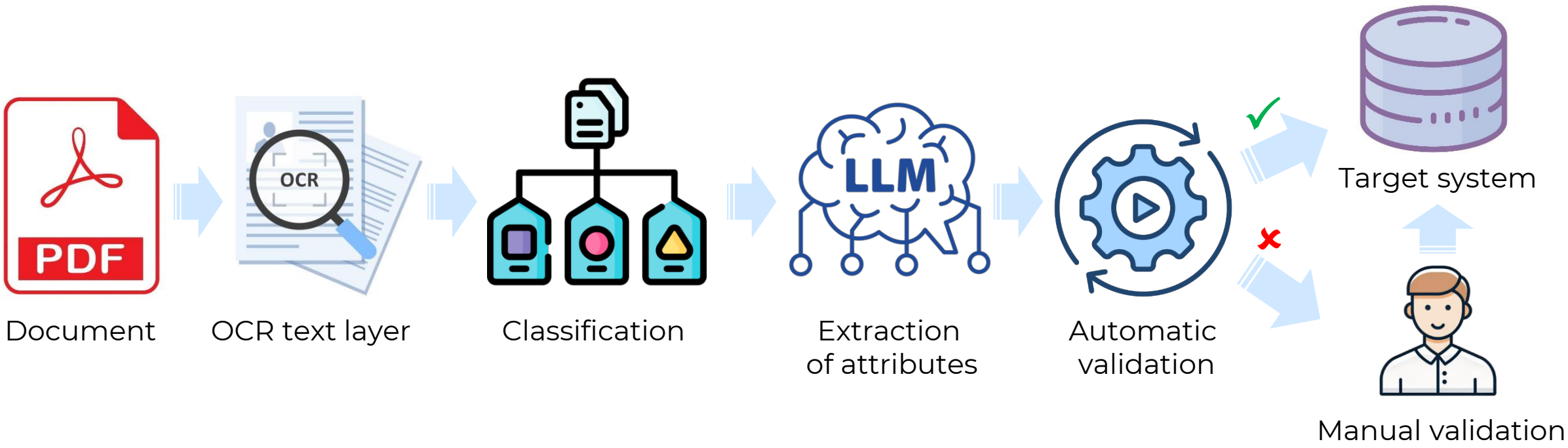
OCR text layer

Classification

Extraction of attributes

Automatic validation

Document processing: reconstruction of the process requires both AI and algorithmic robotization



Document

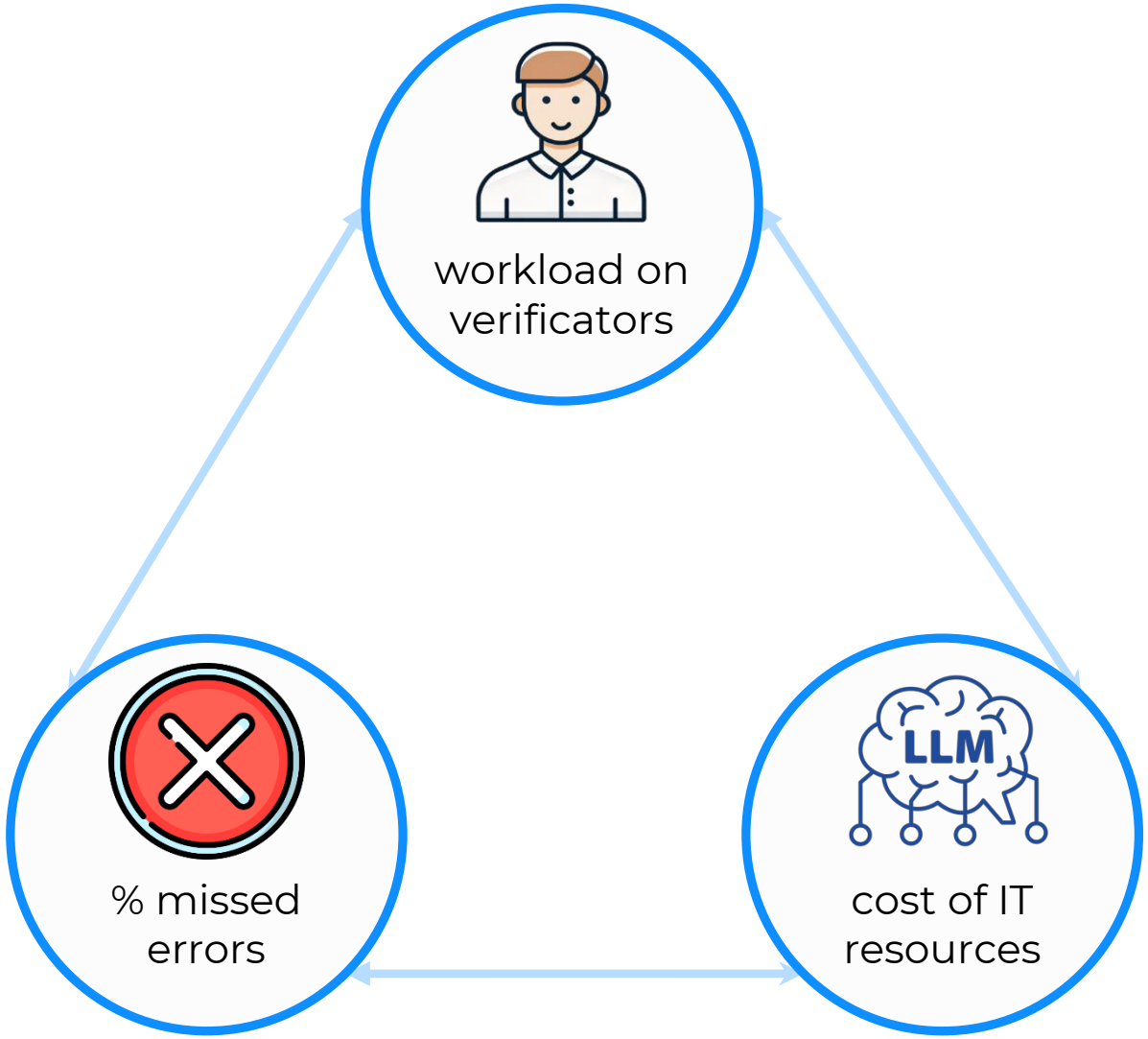
OCR text layer

Classification

Extraction of attributes

Automatic validation

Document processing: trade-off



Challenge:

- Reconstruct the whole process to achieve maximum efficiency
- Propose automatic validation procedures that ensure the number of missed errors does not exceed the number of errors in manual process

  Need to take into account that there are 2 types of errors: related to OCR and LLM
- For some attributes LLM may be not the best choice (sometimes it makes sense to use ML or regular expressions)
- LLM used should be relevant to the task complexity

Preparation of answers to various types of external requests



External request

Preparation of answers to various types of external requests



External request



OCR text layer

Preparation of answers to various types of external requests



External request



OCR text layer



Read the request

- ✓ Marks which standard elements are being requested
- ✓ Determines whether the request is standard or not
- ✓ Writes a hint for the employee regarding non-standard requests

Standard request

Состав запроса (чек-лист)	
1. Сведения о счетах, вкладах, банковских картах	Нет
2. Выписки по расчетным счетам	Да
3. Сведения об остатке денежных средств	Да
4. Содержание картотеки по счету	Нет
5. Сведения об арестах/ приостановлениях операций	Нет
6. Сведения об исполнении инкассовых поручений	Нет
7. Сведения о выданных кредитах и исполнении обязательств	Нет
8. Сведения об имуществе в залоге	Нет
9. Сведения о решениях о приостановлении операций	Нет
10. Копии анкет-заявлений на кредиты	Нет
11. Документы, подтверждающие обоснованность требований кредитора	Да
12. Кредитное досье	Нет
13. Документы, предоставленные должником при заключении договора	Да
14. График платежей	Нет
15. Расчет задолженности	Нет
16. Иные документы, подтверждающие обоснованность требований	Да
17. Банковские реквизиты для погашения задолженности	Нет
18. Запрос на Блокировку операций по счетам и банковским картам	Да
19. Справки о блокировке операций	Нет
20. Запрос на отключение от интернет-банка	Нет
21. Запрос на непроведение исполнения инкассовых поручений без уведомления финансового управляющего	Нет
22. Запрос на непроведение безакцептного списания денежных средств	Нет
Информация по выписке	
Информация из запроса	
Вид запроса	Типовой

Non-standard request

Тригеры для нетипового запроса	
Запрос на перевод средств с указанием конкретных счетов	Нет
Номера счетов банкрота	
Период выписки больше 3 лет от сегодняшней даты	Да
Срок между датой начала выписки и датой признания банкротом больше 3,5 лет	
Наличие запроса по третьим лицам	Нет
Запрос информации прочее (дополнительно к пунктам чек-листа ниже)	Сведения о криптоактивах должника
Состав запроса (чек-лист)	
Информация по выписке	
Информация из запроса	
Вид запроса	Нетиповой

Preparation of answers to various types of external requests



External request



OCR text layer



Read the request

Preparation of answers to various types of external requests



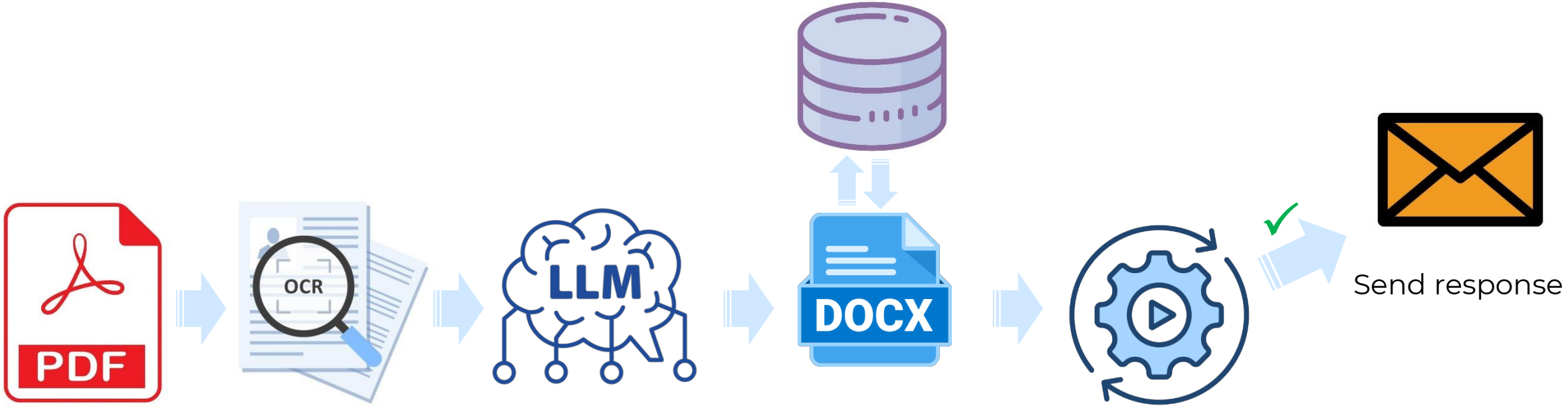
External request

OCR text layer

Read the request

Prepare a standard response

Preparation of answers to various types of external requests



External request

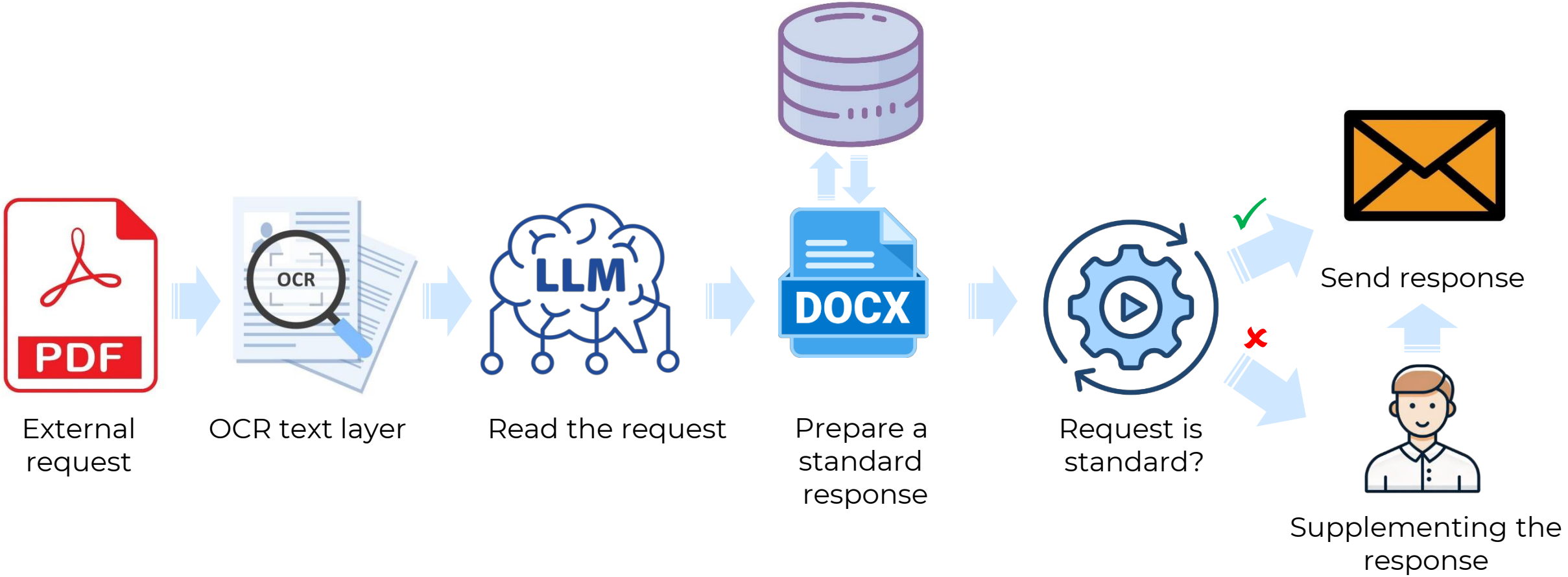
OCR text layer

Read the request

Prepare a standard response

Request is standard?

Preparation of answers to various types of external requests



External request

OCR text layer

Read the request

Prepare a standard response

Request is standard?

Send response

Supplementing the response

Development of AI consultants usually requires application of RAG

Challenge:

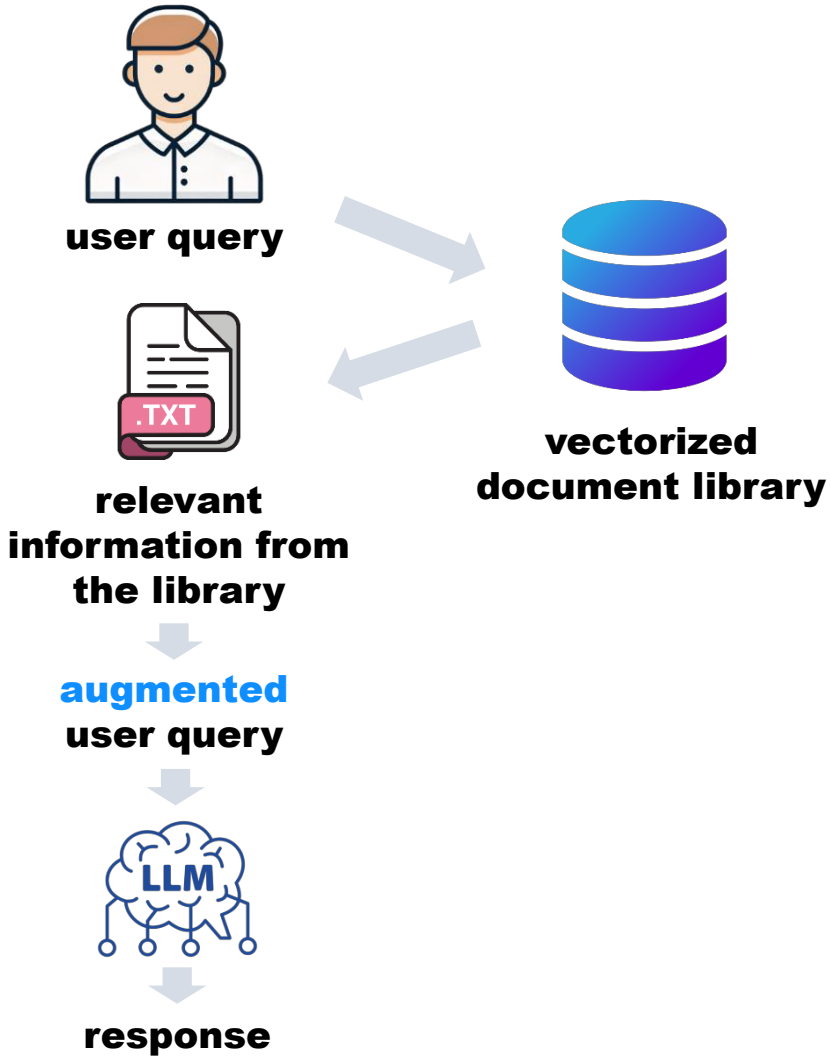
- LLMs only operate with information available at the time of training
- Companies need supplement the models' knowledge base with information relevant at the moment of the user's query and with internal data

Examples:

- Release of a new law or new regulatory requirements
- Changes in software functionality

Solution:

- Create a knowledge library that is continuously updated with relevant information
- Select fragments (chunks) from the knowledge library related to the user's query and pass them to the LLM along with the query



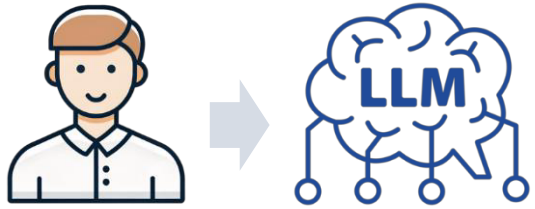
Application: work spaces for methodological queries in accounting, tax, actuarial area as well as for IT support queries are successfully implemented

In practice RAG system requires more complex architecture...



user query

In practice RAG system requires more complex architecture...



user query

variations of user query

query – v.1

query – v.2

query – v.3

In practice RAG system requires more complex architecture...



user query

variations of user query

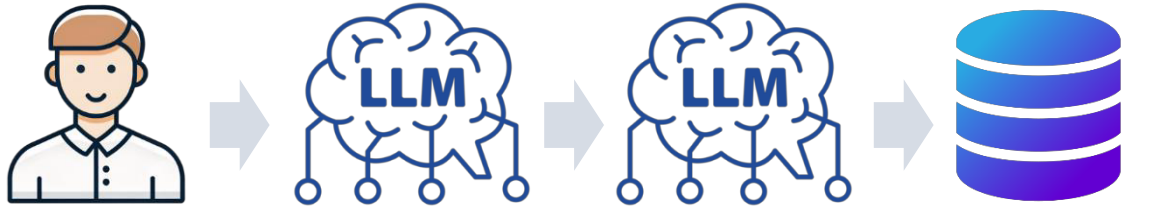
exclude non-meaningful words

query – v.1

query – v.2

query – v.3

In practice RAG system requires more complex architecture...



user query

variations of user query

exclude non-meaningful words

vectorized document library

query – v.1

query – v.2

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In practice RAG system requires more complex architecture...



user query

variations of user query

- query – v.1
- query – v.2
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exclude non-meaningful words

vectorized document library

relevant information from the library

- chunk – 1
- chunk – 2
- chunk – 3
- chunk – 4
- chunk – 5
- chunk – 6

In practice RAG system requires more complex architecture...



user query



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exclude non-meaningful words



vectorized document library



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reranker (e.g. LLM)

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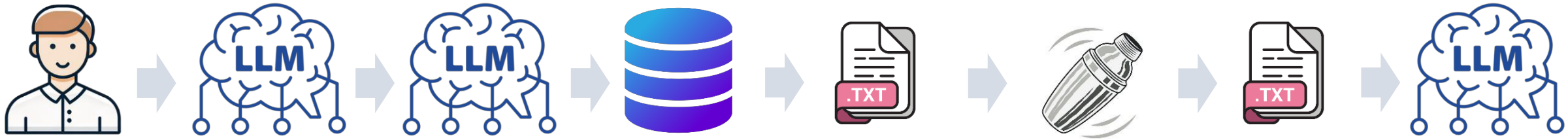
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- chunk – 1
- chunk – 3
- chunk – 5

augmented user query

- user query
- chunk – 1
- chunk – 3
- chunk – 5

In practice RAG system requires more complex architecture...



user query

variations of user query

- query – v.1
- query – v.2
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exclude non-meaningful words

vectorized document library

relevant information from the library

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reranker (e.g. LLM)

- chunk – 1
- chunk – 3
- chunk – 5

augmented user query

- user query
- chunk – 1
- chunk – 3
- chunk – 5

response

Agents that review and update internal documents when legal amendments are introduced could be the next step for RAG technology implementation



Trigger: legislation amendments



Vectorized search of internal documents affected by the amendment





Analysis of changes and preparation of drafts





Review by responsible employees



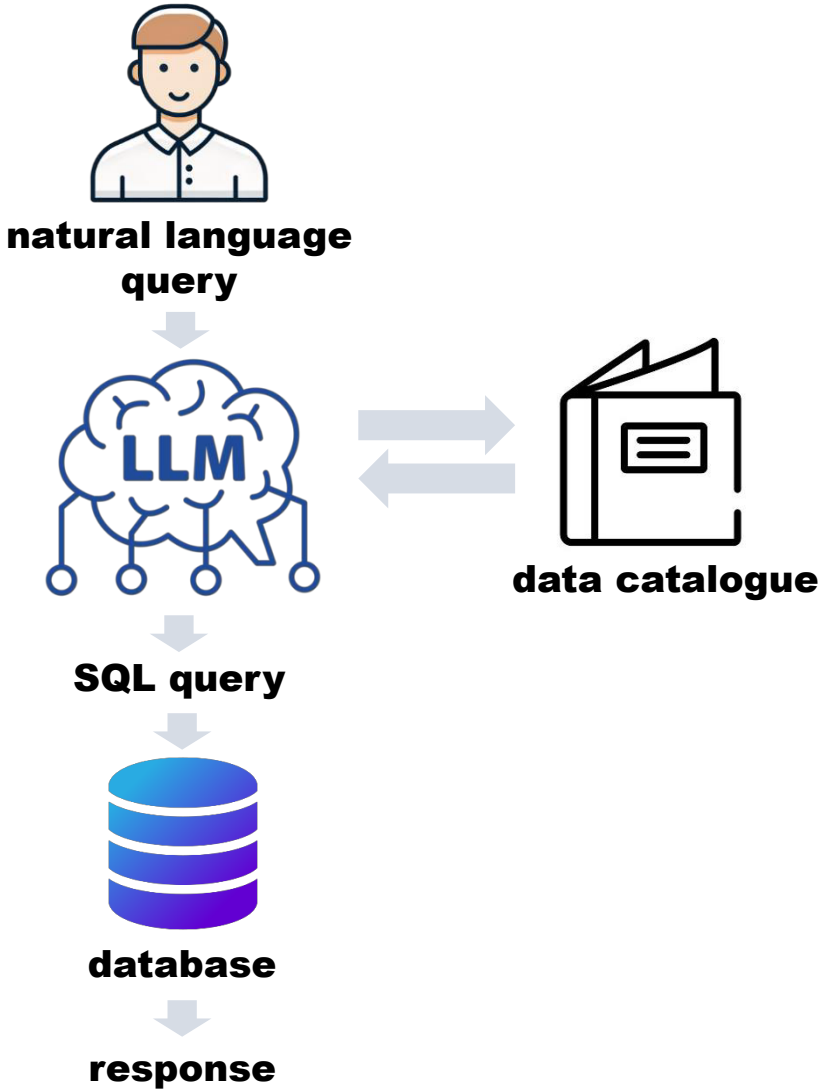
Generation of SQL queries (Text-2-SQL)

Challenge:

- Extraction of data from internal databases requires involvement of analytics
- An employee may urgently need the data at the moment (during a meeting or call)

Solution:

- Prepare a high-quality and consistent description of the attributes contained in the database
- Enrich models' knowledge base with the company's internal terms and requirements
- Use few-shot example technique
- Based on the user's query and the attribute description, use an LLM to generate an SQL query and return a response containing the required information from the database.



Application: retrieving information from databases by employees who do not work with SQL or are not familiar with the data structure in the database

Transaction or balance sheet analysis

Challenge:

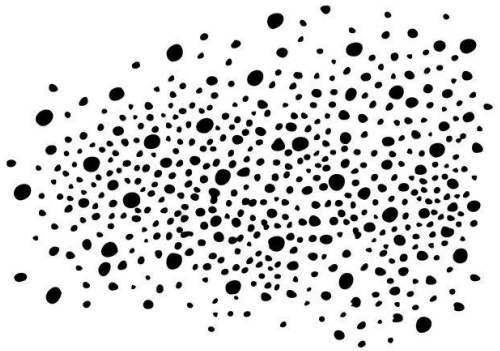
- Need to process huge volumes of transactions
- Processing all transactions with a deterministic algorithm is impossible:
  - many irregular cases that can not be standardized;
  - new scenarios regularly appear.
- Processing all transactions through LLM is impossible:
  - slow processing speed.

Examples:

- Determining the VAT amount based on the payment purpose
- Identifying suspicious transactions
- Checking correctness of postings

Solution:

- Combination of a deterministic algorithm and LLM
- LLM operates with cases that are difficult to handle algorithmically



Deterministic algorithm
95-99%
of transactions/criteria

LLM
1-5%
of transactions/criteria

в т.ч. НДС(20%) 38928-00

Включая НДС 5%, 19801 руб. 91 коп.

ОПЛАТА ПО СЧЕТУ № 123 ОТ 01.01.2025. В Т.Ч. НДС 12900000215000012900000 112900000.00 РУБ. (Сумма платежа: 129 000)

Application: automation of data extraction from transactions; checking transactions/clients for signs of suspicious activity, identify incorrect postings and ensure clear balance sheet